



Digitalization of the Circular Economy in the European Union: Climate Mitigation Potential and Systemic Risks



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Abstract: The integration of digital technologies into circular economy systems has been increasingly promoted within the European Union as a pathway for accelerating climate change mitigation and resource efficiency. However, the environmental implications of large-scale digitalization remain insufficiently understood. In this study, the mitigation potential and systemic risks associated with digital technologies in circular economy transitions were systematically evaluated. A systematic literature review following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses framework was conducted, through which 113 peer-reviewed articles and policy documents published between 2018 and 2025 were synthesized. The results indicate that certain digital technologies—particularly predictive maintenance systems, digital twins, and product life-cycle monitoring platforms—consistently generate net-positive mitigation outcomes by reducing material throughput. In contrast, the mitigation potential of highly data-intensive technologies, including artificial intelligence models and blockchain infrastructures, is frequently constrained by rebound dynamics and the substantial energy requirements of computational infrastructures. Evidence further suggests that efficiency gains achieved through digitally optimized industrial processes are often partially offset by the carbon footprint associated with large-scale model training and the growing electricity demand of digital infrastructures. By integrating technological, environmental, and governance perspectives, a governance-centered analytical framework was proposed to differentiate between net-positive digital circular economy interventions and those whose environmental performance remains conditional on regulatory oversight. The synthesis further reveals significant policy fragmentation between the European Union’s green transition agenda and its digital transformation strategies. These findings highlight the necessity of coordinated regulatory frameworks to ensure that digitalization contributes substantively to circular economy-driven decarbonization.

Keywords: Circular economy; Digitalization; Climate change; Digital technologies; Rebound effects; European Union

1. Introduction

The convergence of the circular economy and digital technologies presents an emergent frontier in climate change mitigation policy and research. The circular economy framework aims to decouple economic growth from resource consumption through strategies such as reuse, recycling, remanufacturing, and product-life extension (Geissdoerfer et al., 2017). Simultaneously, digital technologies—including artificial intelligence, blockchain, the Internet of Things, and big data—are reshaping how production systems, resource flows, and environmental performance are monitored and managed (Bressanelli et al., 2018; Pappas et al., 2018). The European Union has made the twin transition—green and digital—a central priority under the European Green Deal and the digital strategy agenda (European Commission, 2024b; European Commission, 2019). Through instruments such as the Circular Economy Action Plan and the Ecodesign for Sustainable Products Regulation, the European Union increasingly links circular economy implementation with digital infrastructures, traceability tools, and product information systems (European Commission, 2020; European Union, 2024b).

However, this convergence remains ambivalent: digital tools may optimize circular economy processes, yet they can also intensify energy demand, generate rebound effects, and accelerate electronic waste streams (Freitag et al.,

2021; Lange et al., 2020). Climate gains are therefore not automatic; they depend on governance structures, energy systems, and the broader socio-technical context in which digital solutions are deployed (Parrique et al., 2019). Although scholarly interest in this nexus has expanded, much of the literature still privileges enabling narratives or technology-specific applications, while offering less systematic assessment of climate-related trade-offs and governance risks (Chauhan et al., 2022).

This study addresses that gap through a systematic literature review, identifying how digital technologies enable circular economy-based mitigation while also introducing climate-related risks. It focuses on both global perspectives and the European Union's regulatory landscape. The following research questions guide the study.

1. How do digital technologies facilitate circular economy strategies for climate mitigation?
2. What climate-related risks emerge from digital integration in circular economy practices?
3. How do European Union climate and digital policies shape this nexus, and where are the governance gaps?

This study makes several distinct contributions to the literature. First, it reframes the relationship between digital technologies and the circular economy through a climate mitigation and governance lens, moving beyond predominantly techno-optimistic narratives. Second, it systematically distinguishes between digital circular economy practices with unconditional climate benefits and those whose mitigation potential is contingent upon governance, energy sources, and regulatory design. Third, by synthesizing environmental, digital governance, and climate policy studies, the study conceptualizes digitalization not only as an enabler of circularity but also as a source of systemic climate risk. Finally, focusing on the European Union, the study provides a structured assessment of policy coherence between circular economy strategies and emerging digital regulations, highlighting institutional fragmentation and governance gaps that constrain effective climate action.

Despite the growing body of literature on circular economy implementation and digital transformation, existing studies remain analytically fragmented. A significant portion of the literature examines digital technologies primarily as technical enablers of resource efficiency, while circular economy research largely focuses on material flows, waste reduction, and business models without systematically incorporating the climate implications of digital infrastructures. More critically, current research rarely evaluates the net climate mitigation outcomes of digitalized circular economy practices, often overlooking rebound effects, energy intensity, and systemic risks embedded in digital systems. Furthermore, governance-oriented analyses addressing how environmental and digital regulatory regimes interact—particularly within the European Union—remain limited. As a result, there is a lack of an integrated analytical framework that simultaneously captures technological opportunities, climate-related risks, and governance challenges arising from the digitalization of the circular economy. This study addresses this gap by adopting a governance-centered perspective that explicitly links digital circular economy interventions to climate mitigation outcomes and systemic risk dynamics.

2. Methodology

This study employs a systematic literature review to explore the intersection of circular economy, digital technology, and climate change mitigation. The systematic literature review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses framework (Moher et al., 2009) to ensure transparency and replicability. The objective is to synthesize empirical, theoretical, and policy-relevant literature that addresses the digital–circular–climate nexus. The literature search was conducted across the following databases: Scopus, Web of Science, Google Scholar, and ScienceDirect. Policy documents were sourced from the European Commission, European Environment Agency, Organization for Economic Co-operation and Development, and Ellen MacArthur Foundation.

Search terms included combinations of:

1. “circular economy”, “digital technologies” and “climate change mitigation”.
2. “green digital transition” and “emissions reduction”.
3. “Internet of Things,” “artificial intelligence,” “blockchain,” “big data”, and “resource efficiency” or “carbon footprint”.

The search was limited to articles published between 2018 and 2025, peer-reviewed journals, and official policy reports. Grey literature was included only if issued by credible institutions. Studies were included if empirical analyses, theoretical frameworks, European Union policy evaluations, or interdisciplinary reports addressing the intersection of circular economy, digital technologies, and climate change mitigation were presented. In contrast, studies focusing exclusively on purely technical information and communication technology developments without relevance to environmental outcomes were excluded. Publications examining circular economy practices unrelated to climate change mitigation, as well as studies addressing digitalization that did not involve resource use or emissions, were also excluded.

From 215 documents initially retrieved, the following screening steps were performed to improve transparency.

1. Identification and Duplicates: Initial records were screened for duplicates, leading to the exclusion of 42 duplicate records.
2. Title and Abstract Screening: The remaining 173 records were screened, resulting in the exclusion of 60

articles that did not meet the thematic inclusion criteria (e.g., focus on purely technical performance without climate relevance).

3. Full-Text Evaluation: The final 113 sources were evaluated against specific criteria: alignment with digital-circular synergies, evidence of climate mitigation impact or risk, and relevance to European Union policy frameworks.

Following the process (Figure 1), 113 sources were selected for the final analysis. These were coded using thematic analysis (Braun & Clarke, 2006), identifying opportunities and risks under five major themes.

1. Digital technologies enabling circular economy-driven mitigation.
2. Climate risks from digital expansion.
3. Rebound effects and system lock-ins.
4. Governance and regulatory integration.
5. Equity and justice considerations.

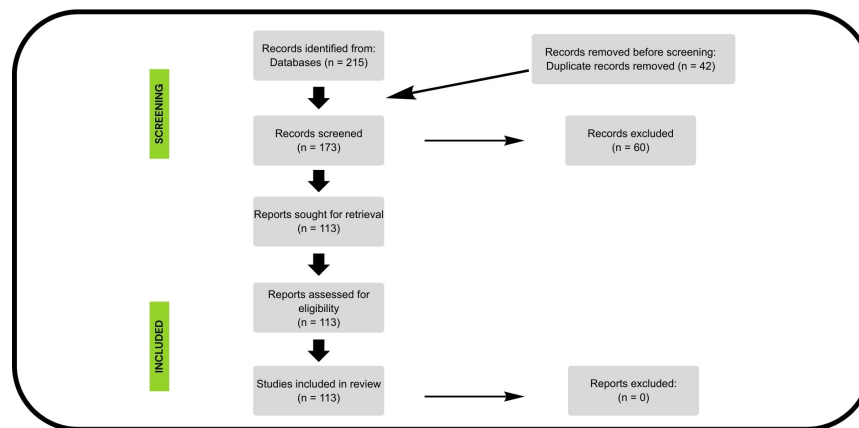


Figure 1. Flow diagram of the study selection process

The thematic analysis followed an inductive coding strategy, allowing themes to emerge from the literature rather than imposing predefined categories. Initial open coding was conducted to identify recurring concepts related to digital technologies, circular economy practices, climate mitigation outcomes, and governance mechanisms. These codes were subsequently clustered into higher-order themes through iterative comparison and refinement. Particular attention was given to studies reporting contradictory findings, especially regarding rebound effects and energy consumption, which were retained to avoid confirmation bias. The resulting thematic structure reflects both converging and diverging perspectives within the literature, thereby enhancing analytical depth and replicability.

3. Digitalization and Circular Economy Pathways for Climate Change Mitigation: Opportunities and Risks

The relationship between digital technologies and circular economy strategies has increasingly been positioned as a central component of contemporary climate mitigation efforts. Through the integration of advanced digital infrastructures, new opportunities for improving resource efficiency, extending product life cycles, and enhancing transparency across value chains have been identified. However, alongside these opportunities, a range of systemic environmental and governance challenges has also emerged. The following analysis differentiates between enabling mechanisms that strengthen circular climate mitigation and structural risks that may undermine these benefits.

3.1 Opportunities: Digital Technologies as Enablers of Circular Climate Mitigation

Digital technologies offer multi-layered opportunities for transforming circular economy strategies into effective tools for climate change mitigation. These opportunities span a broad spectrum—from optimizing product life cycles and monitoring resource flows to tracking real-time emissions and the diffusion of sustainable business models.

Figure 2 categorizes the primary digital drivers of circularity and illustrates their specific functional contributions to climate mitigation. Artificial intelligence analytics acts as the intelligence layer by supporting anticipatory decision-making, while the Internet of Things and digital twins create operational visibility across product use and production processes. Sharing platforms support access-based models that may reduce demand

for new products, and digital traceability tools improve information continuity across value chains. Together, these elements indicate how digitalization can shift circularity from a reactive waste-management logic toward a more proactive and data-driven model. To begin with, Internet of Things-enabled monitoring and sensor systems can help extend product lifespans by identifying maintenance needs and supporting predictive servicing, thereby reducing premature replacement (Rejeb et al., 2022). Likewise, digital twins can improve circular decision-making by simulating product and process performance, reducing material losses, and improving operational efficiency across manufacturing systems (Preut et al., 2021).

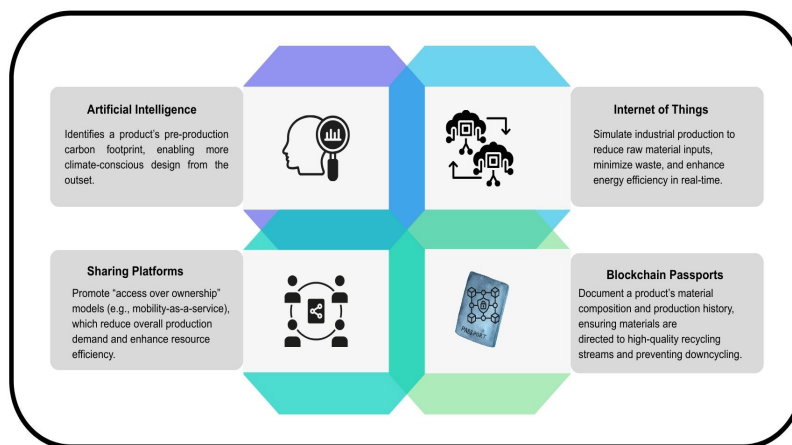


Figure 2. Digital technologies enabling the circular economy

Artificial intelligence and big data analytics facilitate the identification of emission-intensive stages in production and resource use, thereby supporting process redesign and more climate-conscious decision-making (Bag et al., 2021). Blockchain-based traceability systems, particularly when operationalized through digital product passports, can preserve information on material composition, production history, and product status across value chains, which in turn supports repair, reuse, and higher-quality recycling (Upadhyay et al., 2021). Digitalization also enables service-oriented and product-service business models by improving monitoring, maintenance, and coordination across use phases, thereby strengthening circular value retention strategies (Bressanelli et al., 2018; Chauhan et al., 2022).

At the European level, this integration is explicitly endorsed. The Circular Economy Action Plan identifies digital solutions as strategic enablers of circularity, while the Ecodesign for Sustainable Products Regulation provides the regulatory basis for expanding product-related environmental information and digital traceability across the internal market (European Commission, 2020; European Union, 2024b). In this respect, digital product passport architectures have become central to the European Union's effort to align resource efficiency, transparency, and climate-oriented product governance.

The circular model shown in Figure 3 visually synthesizes the iterative relationship between digital intervention and the product life cycle. It illustrates a continuous feedback loop. Extending product life through Internet of Things-driven predictive maintenance reduces the necessity for new resource extraction. This process subsequently contributes to the optimization of material and energy flows, where artificial intelligence substitution strategies lower the emission intensity of the supply chain. The cycle is secured by ensuring transparency and trust, using blockchain to facilitate high-quality recycling at the end of the life cycle. This figure supports the central argument that mitigation benefits are maximized when digital tools are integrated at every stage of the circular process rather than applied as isolated technological solutions. In conclusion, digital technologies have the potential to transform the circular economy from a narrow focus on waste management into a real-time, data-driven, and systemic climate solution. However, realizing this potential requires that digital infrastructures be designed in accordance with sustainability principles and be supported by inclusive policies that ensure equitable access and participation.

The analysis reveals that digital technologies differ substantially in terms of their climate mitigation performance within circular economy systems. While applications such as predictive maintenance, digital twins, and product life-cycle monitoring generally exhibit net-positive mitigation effects, other technologies—particularly data-intensive artificial intelligence applications and blockchain-based systems—produce conditional outcomes that depend on energy sources, scale of deployment, and regulatory constraints. In the absence of targeted governance mechanisms, these technologies risk amplifying emissions through rebound effects and infrastructure-related energy demand. This distinction underscores the importance of policy design in determining whether digitalization supports or undermines circular climate objectives.

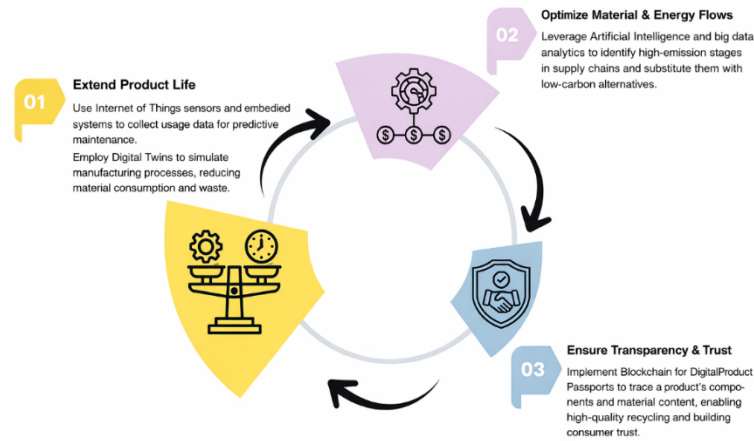


Figure 3. Data-driven circular economy model

3.2 Risks: Rebound Effects, Energy Demand, and Digital Waste

While digital technologies offer promising avenues to advance circular economy strategies for climate change mitigation, they also introduce substantial environmental and systemic risks. These risks often emerge not from the direct use of the technologies themselves, but from their secondary and indirect impacts. Among the most significant of these are rebound effects, increasing energy consumption, and the proliferation of electronic waste.

Figure 4 quantifies the systemic carbon burden associated with the twin transition and highlights the material and energetic costs embedded in digitalization. Even when digital applications improve process efficiency at the firm level, infrastructure-level electricity demand and rebound dynamics can erode or partially offset these gains. The rebound effect refers to situations in which efficiency improvements lower costs and thereby stimulate additional consumption or throughput (Gossart, 2014). In this context, climate benefits may be weakened when digitally enabled efficiencies are accompanied by scale expansion or additional demand, a dynamic widely discussed in the broader sustainability literature as a structural limitation of efficiency-led strategies (Parrique et al., 2019). Shared or platform-based services may also produce mixed outcomes when they substitute low-carbon practices in one context but stimulate new consumption in another (Zink & Geyer, 2017).

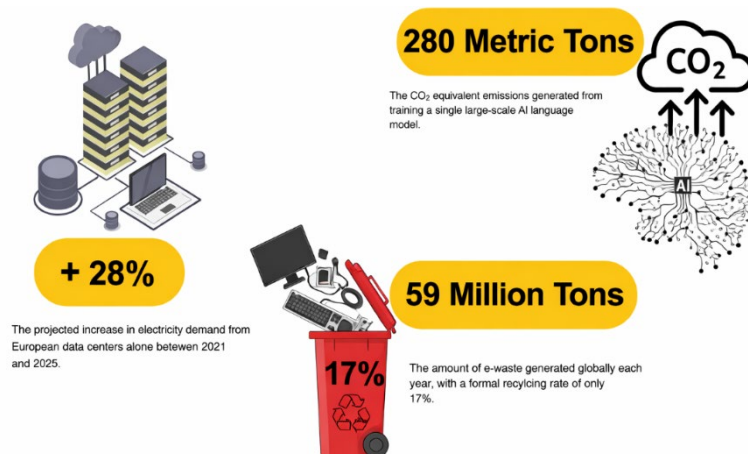


Figure 4. Energy and emission impacts of artificial intelligence, data centers, and digital waste

A second major risk concerns the energy intensity of digital infrastructures. Data centres, artificial intelligence models, blockchain systems, and next-generation network infrastructures require substantial electricity inputs, which complicates claims that digitalization is environmentally benign by default (Lange et al., 2020). Recent assessments by the International Energy Agency indicate that the rapid expansion of artificial intelligence is becoming an important driver of rising data-centre electricity demand, thereby increasing the likelihood that efficiency gains in downstream applications will be offset at the infrastructure level (International Energy Agency, 2025). In addition, model development itself can be highly carbon-intensive: Strubell et al. (2019) show that the training of large natural language processing models can generate substantial greenhouse gas emissions.

Thirdly, the widespread diffusion of connected devices accelerates electronic waste generation. The rapid

turnover of smartphones, sensors, and other digitally enabled products shortens effective use cycles and expands disposal volumes, even where operational efficiency improves. According to The Global E-waste Monitor 2024, global electronic waste reached 62 million tonnes in 2022 and is projected to rise to 82 million tonnes by 2030, while only 22.3% was formally collected and recycled (Baldé et al., 2024). From a circular economy perspective, this trend is especially problematic because low collection rates hinder the recovery of critical materials and transfer environmental burdens to downstream waste systems.

Beyond energy use and waste generation, digital systems can also externalize environmental pressures across global supply chains—from mineral extraction and hardware manufacturing to disposal and end-of-life treatment. This broader life-cycle perspective is essential because the environmental footprint of digitalization is not confined to the point of use; rather, it is distributed across infrastructures, labor regimes, and material extraction chains that frequently remain invisible in techno-optimistic policy narratives (Crawford & Joler, 2019; Freitag et al., 2021).

Global electronic waste reached 62 million tonnes in 2022 and is expected to continue rising under conditions of accelerated hardware turnover and expanding digital infrastructures (Baldé et al., 2024). Recent research has also shown that the expansion of generative artificial intelligence may intensify this pressure by increasing demand for specialized servers and shortening equipment replacement cycles, potentially adding substantial volumes of electronic waste over time (Wang et al., 2024). For digital tools to support circularity in substantive rather than symbolic terms, design principles must therefore extend beyond technical efficiency to include durability, repairability, material recovery, and infrastructure governance.

4. Governance and European Union Policy Landscape

From a governance perspective, the digitalization of the circular economy represents a complex socio-technical system characterized by cross-sectoral interactions, regulatory overlap, and risk externalization. Drawing on policy-coherence and governance perspectives, effective climate mitigation requires alignment between environmental regulation, digital policy, and industrial strategy. In the absence of such alignment, efficiency gains at the micro level may coexist with unintended macro-level environmental risks. Within the European Green Deal, the Circular Economy Action Plan encourages systemic changes across the product life cycle, while the Ecodesign for Sustainable Products Regulation strengthens the legal foundation for digital product information and traceability across markets (European Commission, 2020; European Union, 2024b). These instruments expand the role of data infrastructures in decisions related to reparability, recyclability, material composition, and product stewardship.

Nonetheless, the digital dimension of this transformation continues to operate within a complex regulatory environment. The European Union's broader digital rulebook increasingly addresses issues of data governance, accountability, and system oversight, and the Artificial Intelligence Act adds a more explicit layer of regulatory control for artificial intelligence systems (European Union, 2024a). Even so, the integration of digital regulation with circular economy implementation remains incomplete. A related challenge concerns interoperability and shared governance arrangements. The Commission's work on common European data spaces points to the importance of secure and interoperable data-sharing environments, yet their practical integration into circular economy governance is still evolving (European Commission, 2024a).

A further governance challenge lies in uneven implementation capacity across member states. Countries with stronger digital infrastructures, higher administrative capacity, and more mature industrial ecosystems are better positioned to operationalize digital-circular integration, while others may lag behind. This asymmetry risks reinforcing territorial disparities in the benefits of the twin transition. In addition, limited data sharing, commercial sensitivity, and trust deficits across value chains continue to constrain the information flows on which circular digital systems depend (Upadhyay et al., 2021).

Ultimately, policy coherence between environmental and digital frameworks is essential if digital transformation is to produce durable climate benefits. When circular economy regulation and digital governance evolve as separate policy streams, climate-relevant risks embedded in data infrastructures, software-intensive services, and hardware turnover are less likely to be addressed in a coordinated manner. For this reason, future European Union strategies should treat digitalization and circularity as mutually conditioning domains rather than parallel agendas (Gritsenko & Wood, 2020).

5. Conclusion and Policy Implications

This study contributes to the academic literature by advancing an integrated analytical perspective on the digitalization of the circular economy for climate change mitigation. Unlike existing studies that predominantly emphasize technological potential or sector-specific applications, this study demonstrates that the climate effectiveness of digital circular economy strategies is inherently contingent upon governance structures, energy systems, and regulatory coherence. By conceptualizing digitalization as both a facilitator of circular mitigation and a generator of systemic climate risk, the study bridges previously disconnected strands of research on the circular

economy, digital governance, and climate policy. In doing so, it provides a governance-oriented framework that can inform both future empirical research and policy design in the context of the European Union's twin transition. These risks, particularly the energy-intensive nature of digital infrastructures, the rapid growth of electronic waste, and rebound effects, pose significant challenges to net-zero ambitions. Although the European Union's policy ecosystem acknowledges the intersection between digitalization and environmental sustainability, a coherent and operational governance architecture is still lacking.

This study differs from mainstream approaches that view the digitalization of the circular economy merely as a technical optimization process and instead reframes it as a socio-technical governance issue. Much of the existing literature treats digital technologies as neutral inputs that improve resource efficiency. In contrast, this study argues that the net climate mitigation impact of digitalization is shaped primarily by governance structures, energy systems, and the coherence of regulatory frameworks. Earlier contributions, including Jabbour et al. (2018) and Bressanelli et al. (2018), have shown how Industry 4.0 tools can enable circular business models. However, these enabling perspectives only partially capture the systemic risks associated with digital infrastructures themselves, including the emissions implications of data-intensive artificial intelligence, the growing electricity demand of data centres, and the waste pressures associated with accelerated hardware turnover. Addressing this overlooked dimension constitutes one of the key motivations of the present study. The main originality of this study lies in distinguishing between digital circular-economy interventions that generate unconditional climate benefits and those whose climate benefits depend on governance conditions. Technologies such as predictive maintenance, which directly reduce material throughput and resource waste, tend to produce clearer net-positive environmental outcomes. By contrast, data-intensive applications such as artificial intelligence systems or blockchain-based infrastructures can contribute to climate mitigation only when accompanied by low-carbon electricity supply, robust efficiency standards, and regulatory mechanisms capable of containing rebound effects.

Based on the findings of this study, several policy recommendations were proposed to ensure that the digitalization of the circular economy contributes effectively to climate mitigation and resource sustainability.

(a) Mainstream digital sustainability in regulatory frameworks

Digital sustainability must be embedded in all relevant legislative instruments. Digital infrastructures should be evaluated not only on the basis of cybersecurity and economic performance but also on carbon intensity and circular compatibility. Environmental key performance indicators must become standard in procurement, certification, and regulatory assessment.

(b) Carbon taxation or quotas for energy-intensive digital systems

Policy mechanisms such as carbon-based taxation or energy quotas should be introduced for high-emission digital systems. For instance, large artificial intelligence models could be categorized according to their carbon footprint, with additional costs imposed on models exceeding specific thresholds—thereby incentivizing efficiency in algorithm design and model deployment.

(c) Electronic waste governance reform for emerging technologies

Electronic waste policies must be adapted to the proliferation of the Internet of Things devices and smart products with short life cycles. Regulatory mandates should include minimum software support periods, mandatory reparability, and modular hardware standards. Such measures would extend product lifespans and reduce waste generation at the source.

(d) Global application of digital product passports

The Digital Product Passport scheme should be extended beyond intra-European Union production to include imported goods. Without this scope, the environmental digital standard risks becoming a regional construct with limited global influence. A border-adjustment mechanism for environmental data compliance should be considered.

(e) Strengthen policy coherence and cross-sectoral governance

Climate, digital, industrial, and waste governance should not evolve in silos. A unified strategic platform—such as a digital-circular governance forum at the European Union level—can facilitate horizontal integration across directorates and stakeholder groups. This would ensure that digital innovations serve circularity and climate goals in tandem.

In conclusion, the European Union's efforts to digitalize the circular economy should move beyond technological optimism and adopt an approach that places systemic risks and governance requirements at the center of policy design. The efficiency gains provided by digital technologies can translate into meaningful climate mitigation only when the environmental costs of digital infrastructures—particularly rising electricity demand, hardware turnover, and electronic waste generation—are governed effectively. Future research should therefore examine the social and climate justice dimensions of the digital-circular transition and advance standardized carbon accounting approaches for digital infrastructures within circular economy assessment.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflict of interest.

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